



Supply Chains with Python: From Data Simulation to Machine Learning Solutions

Tisha Ghosh
Project Linked Person
ISI, Kolkata



What is Supply Chain Management (SCM)?

Definition:

- Supply Chain Management (SCM) is the coordination and management of all activities involved in sourcing, procurement, conversion, and logistics. It ensures the efficient flow of goods, services, information, and finances across the supply chain.

Key Components:

- **Planning:** Demand forecasting and inventory management.
- **Sourcing:** Supplier selection and relationship management.
- **Production:** Manufacturing and assembling products efficiently.
- **Logistics:** Distribution, warehousing, and transportation.
- **Returns:** Handling product returns and reverse logistics.

What is Supply Chain Management (SCM)?

Objectives:

- Minimize costs.
- Enhance customer satisfaction.
- Improve efficiency and responsiveness.
- Foster collaboration across stakeholders.

Why is SCM Important?

- Reduces operational costs.
- Increases competitiveness.
- Ensures sustainability.
- Meets customer expectations promptly.

The Pizza Delivery Delay

- **The Situation:**

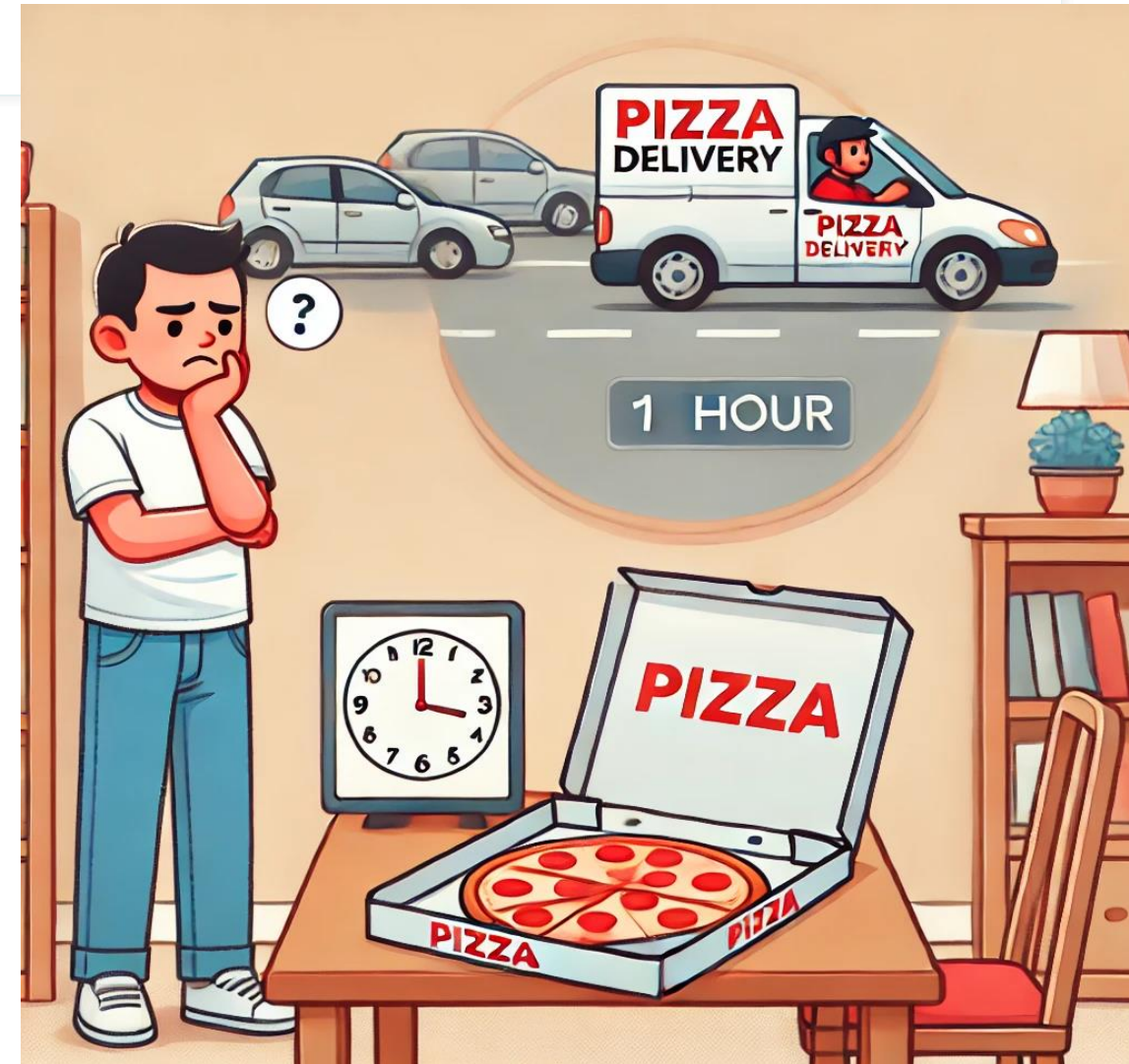
Imagine you order a pizza online. You're told it will arrive in **30 minutes**, but it takes **1 hour** to reach you.

- **What Went Wrong?**

- ✓ The **pizza outlet ran out of ingredients** (poor inventory management).
- ✓ The delivery driver got stuck in **traffic congestion** (logistics issue).
- ✓ Your address was entered **incorrectly in the system** (data error).

- **Impact:**

- ✓ **You're frustrated** and less likely to order again.
- ✓ The pizza outlet offers a discount, increasing **their costs**.
- ✓ **Their reputation suffers** due to negative reviews.



Challenges in Supply Chain Management

Key Problems	Definition	Causes	Impact	Example
1. Inventory Mismanagement	Poor control of inventory levels, leading to stockouts or overstocking.	Inaccurate demand forecasting, lack of inventory tracking systems.	• Lost sales due to stockouts. • Increased holding costs due to overstocking.	A clothing store orders too much winter clothing and ends up with excess stock after the season ends.
2. Bottlenecks	Points in the supply chain where processes are slower, reducing overall throughput.	Insufficient capacity, resource constraints, inefficient workflows.	• Delays in production and delivery. • Increased lead times and customer dissatisfaction.	A factory that lacks enough workers to assemble products causes delays in fulfilling customer orders.
3. Bullwhip effects	The amplification of demand fluctuations as they move up the supply chain.	Miscommunication, over-ordering, lack of demand visibility.	• Overstocking or understocking. • Increased operational costs.	A small increase in customer demand for toilet paper leads retailers to order significantly more from wholesalers, causing manufacturers to ramp up production unnecessarily.

Challenges in Supply Chain Management

Key Problems	Definition	Causes	Impact	Example
4. Supply Chain Disruptions	Interruptions in the flow of goods due to unexpected events.	Natural disasters, geopolitical issues, supplier failures.	• Delayed shipments and production halts. • Loss of revenue and market trust.	A hurricane damages a shipping port, delaying the delivery of imported goods.
5. Inefficient Communication and Data Sharing	Lack of transparency and real-time data exchange across stakeholders.	Siloed operations, outdated technology.	• Increased errors in forecasting and decision-making. • Reduced supply chain agility.	A retailer fails to inform a supplier about a sudden spike in demand, leading to stockouts.
6. Sustainability Challenges	Difficulty in balancing profitability with environmental and social responsibility.	Inefficient processes, waste generation, lack of green logistics.	• Higher regulatory risks. • Damage to brand reputation.	A company faces backlash for using non-recyclable packaging, prompting customers to switch to competitors.

How Can We Improve the Pizza Delivery?

Better Inventory Management	Route Optimization
 Keep ingredients in stock	 Use GPS for optimized routes
Real-Time Traffic Updates	Data Accuracy
 Monitor and adjust routes	 Verify addresses for accuracy

Solutions to Overcome Supply Chain Challenges

Problem	Solution
Inventory Mismanagement	Implement Real-Time Inventory Systems: Use automated software to monitor stock levels and track usage patterns. Adopt Just-In-Time (JIT) Inventory: Align inventory replenishment with actual demand to reduce excess stock. Regular Audits: Conduct periodic stock checks to ensure accuracy and address discrepancies. Demand Forecasting: Predict accurate demand patterns to ensure the right stock levels are maintained, minimizing waste and shortages.
Bullwhip Effect	Improve Demand Forecasting: Use advanced analytics and AI to predict customer demand accurately. Share Real-Time Data: Enhance collaboration by sharing demand data across suppliers and distributors. Reduce Lead Times: Streamline order cycles and production schedules to minimize variability.
Bottlenecks	Identify and Prioritize Bottlenecks: Use process mapping tools like value stream mapping to locate bottlenecks. Increase Capacity: Invest in resources (e.g., machinery, labor) where frequent delays occur. Optimize Scheduling: Stagger production and delivery schedules to avoid overloading critical nodes. Demand Forecasting: Anticipate spikes in demand and allocate resources (e.g., labor, machinery) effectively to prevent bottlenecks.



Solutions to Overcome Supply Chain Challenges

Problem	Solution
Supply Chain Disruption	Build Resilient Supply Chains: Diversify suppliers and build relationships with multiple sources. Adopt Risk Management Plans: Use predictive tools to assess risks and develop contingency plans. Maintain Safety Stock: Keep a buffer of essential inventory to handle emergencies. Demand Forecasting: Identify trends and prepare for demand shifts, enabling proactive measures like safety stock or alternate supplier sourcing.
Inefficient Communication and Data Sharing	Invest in Unified Systems: Implement integrated platforms for real-time communication between all stakeholders. Standardize Data Formats: Ensure all partners use the same data structure to reduce errors. Train Teams on Tools: Educate employees on communication and collaboration tools to improve data sharing. Demand Forecasting: Shared forecasting data can act as a single source of truth, improving collaboration and reducing miscommunication.
Sustainability Challenges	Adopt Green Logistics: Optimize routes to reduce fuel consumption and emissions. Source Sustainably: Partner with eco-friendly suppliers and prioritize recyclable materials. Implement Circular Economy Practices: Promote recycling, refurbishing, and reusing products and materials. Demand Forecasting: Avoid overproduction by aligning production with actual demand, reducing waste and emissions.

Why Demand Forecasting is a Central Solution?

- Every supply chain player works to satisfy the customer.
- **Customer Focus = Reduced Problems**
(Less waste, fewer delays, smoother operations)
- **Meeting Demand = Customer Satisfaction**
(Right products, at the right time)
- Demand forecasting ensures customer happiness and a successful supply chain.



Data: The Foundation of Demand Forecasting

1. What is Data?

- Data is the information we use to predict demand patterns.

2. Why is Data Important?

- Accurate data ensures reliable forecasts, leading to better decisions and customer satisfaction.
- Without good data, forecasting fails, causing delays, waste, and inefficiencies.

3. How Can We Get Data?

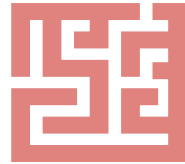
- **1. Secondary Data:** Use existing data from public sources like Kaggle, government databases, or industry reports.
- **2. Simulated Data:** Generate your own data by simulating different scenarios, testing various demand patterns.

What is Simulation?



1. Definition:

"Simulation is the process of creating a model to imitate real-world scenarios and test different conditions."



2. Why Use Simulation?

It allows us to test demand patterns, evaluate supply chain strategies, and predict outcomes without real-world consequences.



3. Key Benefits:

Test Hypotheses: Simulate different market conditions to see how changes affect demand.

Experiment with Scenarios: Try multiple scenarios to identify the best strategy before implementation.

Minimize Risk: Assess potential risks and optimize decision-making without affecting real operations.

Types of Simulation in Supply Chain

Name	Description
1. Discrete Event Simulation (DES)	- Models individual events in a system (e.g., order processing, shipping). - Tracks system performance metrics like lead time, throughput, and bottlenecks.
2. Monte Carlo Simulation	- Uses random sampling to model uncertainty in supply chain parameters (e.g., demand variability, delivery times). - Helps in risk analysis and decision-making under uncertainty.
3. Agent-Based Simulation (ABS)	- Simulates behaviour and interaction of individual agents (e.g., suppliers, manufacturers, retailers). - Focuses on decentralized decision-making and emergent system behaviour.
4. System Dynamics Simulation (SD)	- Models high-level interactions and feedback loops in supply chain networks. - Ideal for analyzing long-term strategic impacts.
5. Hybrid Simulation	- Combines multiple simulation types (e.g., DES and ABS) to address complex supply chain problems. - Provides a comprehensive view of system dynamics and granular details.

What is Discrete Event Simulation (DES)?

Definition:

- Discrete Event Simulation (DES) is a mathematical and computational modeling technique used to analyze and replicate the behavior of complex systems over time. It focuses on discrete events, which are specific occurrences that cause a change in the state of the system. In DES, time progresses in steps from one event to the next, making it an effective tool for studying systems where changes happen at distinct points in time.

Key Characteristics of DES:

- **Event-Based:** Changes occur only when predefined events take place (e.g., a machine starting or a customer arriving).
- **State Changes:** The system's state remains constant between events.
- **Time Advancement:** Time progresses by jumping to the next event, rather than continuously.

Applications of DES:

- **Manufacturing:** To analyze production lines and identify bottlenecks.
- **Healthcare:** To improve patient flow in hospitals.
- **Logistics:** To optimize warehouse operations or transportation systems.
- **Customer Service:** To minimize wait times and improve staffing.

Forecasting Techniques: From Traditional to Machine Learning

- **What is Forecasting?**

Forecasting is the process of predicting future events, trends, or behaviors based on historical data and analysis.

- **How Does Forecasting Work?**



- **Types of Forecasting:**

I. Qualitative Forecasting: Based on subjective judgment, intuition, or expert opinion.

II. Quantitative Forecasting: Based on mathematical models and historical data.

Forecasting Techniques: From Traditional to Machine Learning

Level	Techniques	Strengths	Limitations
Traditional Methods	Moving Average, Linear Regression, Exponential Smoothing, ARIMA	Simple, effective for stable data patterns	struggles with complexity
Machine Learning (ML)	Decision Trees, Random Forest, SVM, Xgboost	Handles complexity	needs more data and computation
Deep Learning (DL)	RNN, LSTM, CNN, Reinforcement Learning (RL)	Best for large, complex data	requires substantial computing power

Data Making: A Practical Example in Python

Q1. Problem Statement:

There is given the task to generate weekly pizza order data for a restaurant using python:

- I. How can you create a dataset where weekly pizza orders increase linearly over time?
- II. How can you simulate pizza orders when a discount is applied?

Solution:

- I. Linear Weekly Pizza Orders:

Formula:

$$\text{Orders} = \text{Base Orders} + \text{Week}$$

- II. Simulating Orders with Discount:

Formula:

$$\text{Orders} = \text{Base Orders} + (10 * \text{Discount value})$$

Then we will use python to make the dataframe.

Inventory Management and Restocking Simulation

Q2. Problem Statement:

You are tasked with simulating the stock levels and sales of a product over a period of time. The product has an initial stock, a reorder point, and a safety stock. The goal is to simulate daily sales for a fixed number of days, track the stock levels, and reorder stock when necessary.

Solution:

Key Formulas Used:

i. **Sale = Random number generate**

ii. **$Stock_{new} = Stock_{old} - Sales_{day}$**

iii. **$Unit\ to\ order = Reorder\ point - Stock_{new} + Sales_{day}$**

Inventory Simulation: Sales, Restocking, and Stock Status Tracking

Q3. You are tasked with simulating the stock levels and sales of multiple products over a given period. The goal is to track daily sales, stock levels, and restocking operations for each product based on specific conditions. Follow these requirements:

- a) Product:
Each product has the following properties: Product ID, Name of the product, Initial stock, Inventory, Reorder point, Safety Stock, Lead time
- b) Simulation Setup: simulate the sales and stock level for 30 days starting from 1/1/2024
- c) Restocking logic: If product stock falls below or equal to the reorder point place a reorder.

Write a Python program that generates a Pandas DataFrame to display the simulated data for all 30 days.

- I. Perform basic EDA.
- II. Plot sales over time.
- III. Draw correlation matrix

Retail Store Demand Simulation and Inventory Management

Q4. Imagine you are the manager of a retail store that sells a popular product (e.g., smartphones, laptops, etc.). Each day, customers arrive and purchase a random number of products. The demand for products varies from day to day, but on average, the store sells 5 units per day. Your goal is to manage the store's inventory and ensure that the store can meet customer demand without running out of stock or overstocking.

The store has the following parameters:

- Initial Inventory:** 50 units.
- Restock Level:** The store will restock when the inventory falls below 20 units.
- Restock Amount:** The store adds 40 units to the inventory whenever restocking is needed.

You need to simulate the following process for **30 days**:

1. Each day, the demand for the product will follow a **Poisson distribution** with an average demand of 5 units per day.
2. Track the store's **inventory** each day.
3. If the inventory falls below the restock level, restock the inventory by 40 units.
4. After simulating for 30 days, visualize the **daily demand** and the **inventory levels** to understand how well the store managed its stock.

Questions:

1. Write a Python simulation using **SimPy** to model the store's demand and inventory management.
2. How does the inventory change over time based on demand? At what points does the store need to restock?
3. Plot the **daily demand** and the **inventory level** over the 30-day simulation period.
4. Use different methods to forecast the future demand.



Leveraging Neural Network Simulations to Analyze and Optimize the Bullwhip Effect in Supply Chains

Agenda



Data Simulation: Creating Realistic Supply Chain Datasets

- **Objective:** Learn how to generate synthetic datasets that resemble real-world supply chain data.



Machine Learning Applications in Supply Chain

- **Objective:** Explore how Machine Learning can solve common supply chain problems.



Building a Basic Machine Learning Model

- **Objective:** Create a simple ML model to solve a supply chain problem.



Hands-On Practice

- **Objective:** Build your own supply chain dataset and ML model.